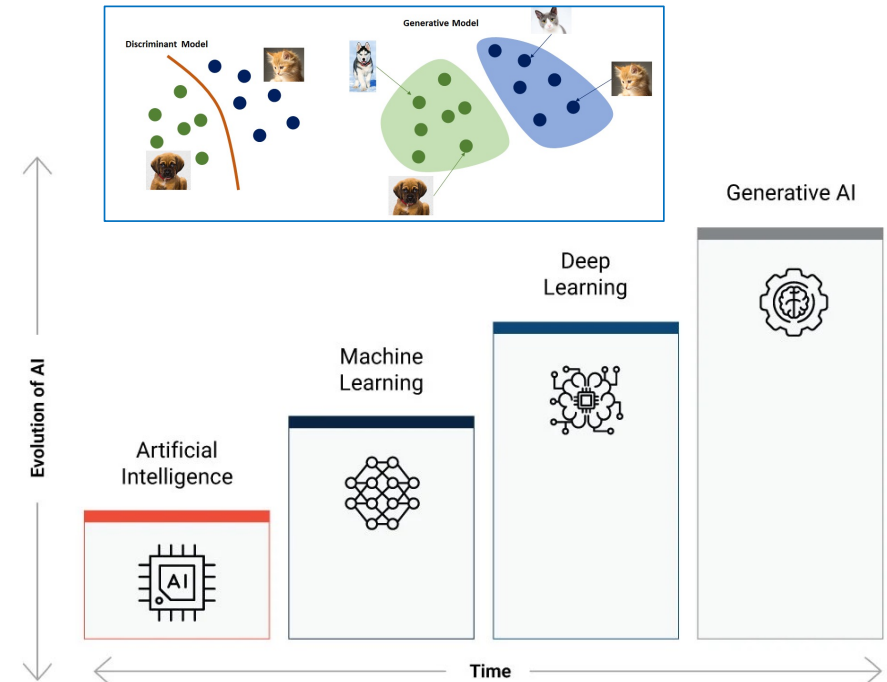
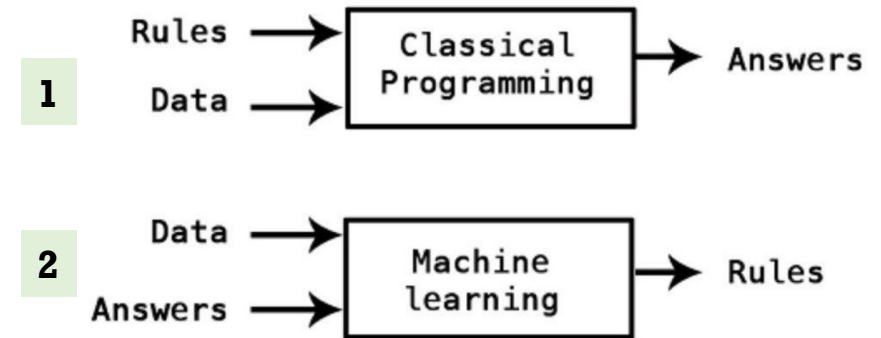
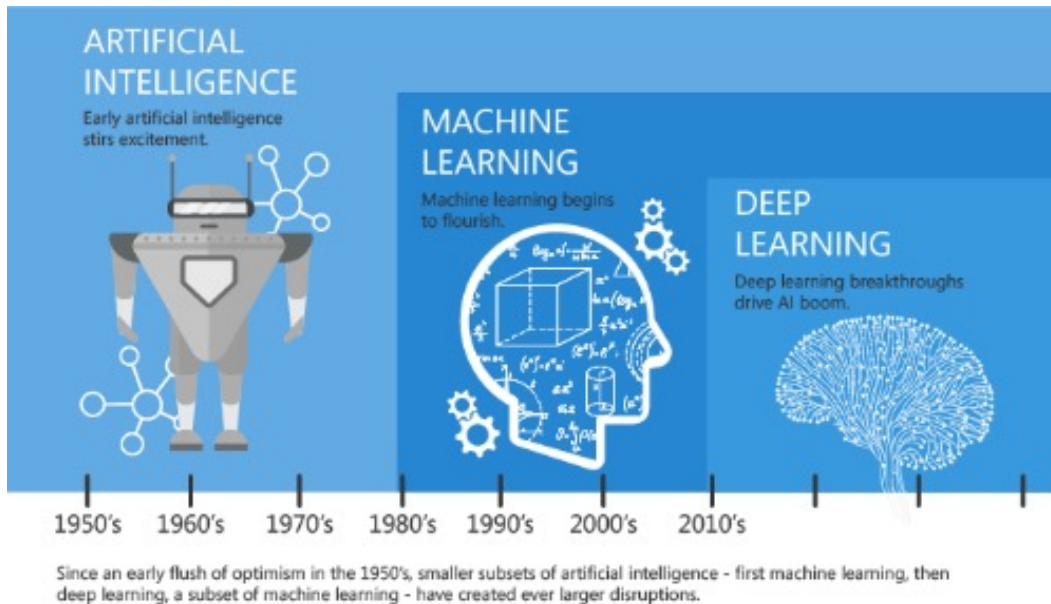




AI in Action!

AI = Automation

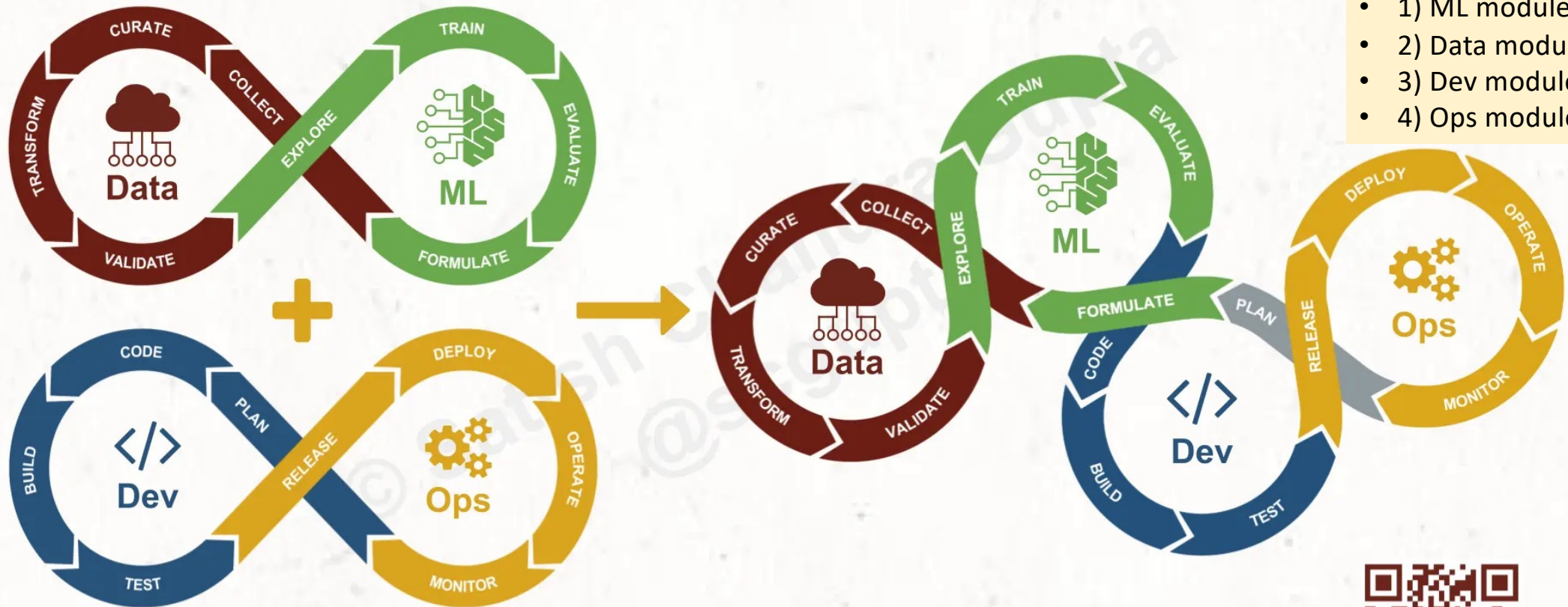
- 1) Rule-based AI
- 2) Machine Learning (ML)



<https://mc.ai/machine-learning-basics-artificial-intelligence-machine-learning-and-deep-learning/>

MLOps = DataML + DevOps

ml4devs.com/mlops-lifecycle 



- 1) ML module
- 2) Data module
- 3) Dev module
- 4) Ops module

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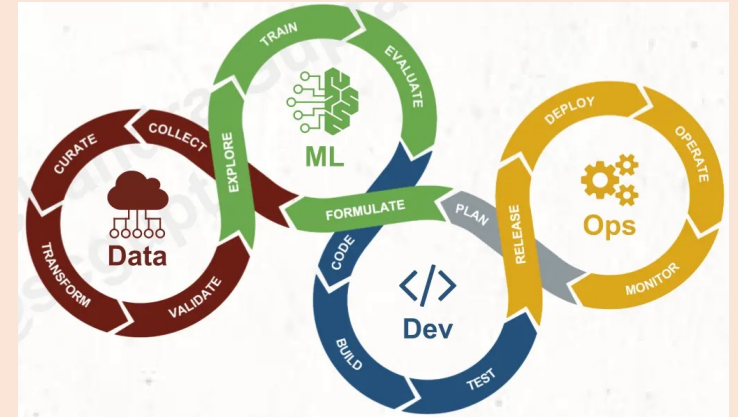
CC BY-NC-ND 4.0 International License

<https://www.ml4devs.com/images/illustrations/ml-lifecycle-fusing-model-and-software-development.webp>

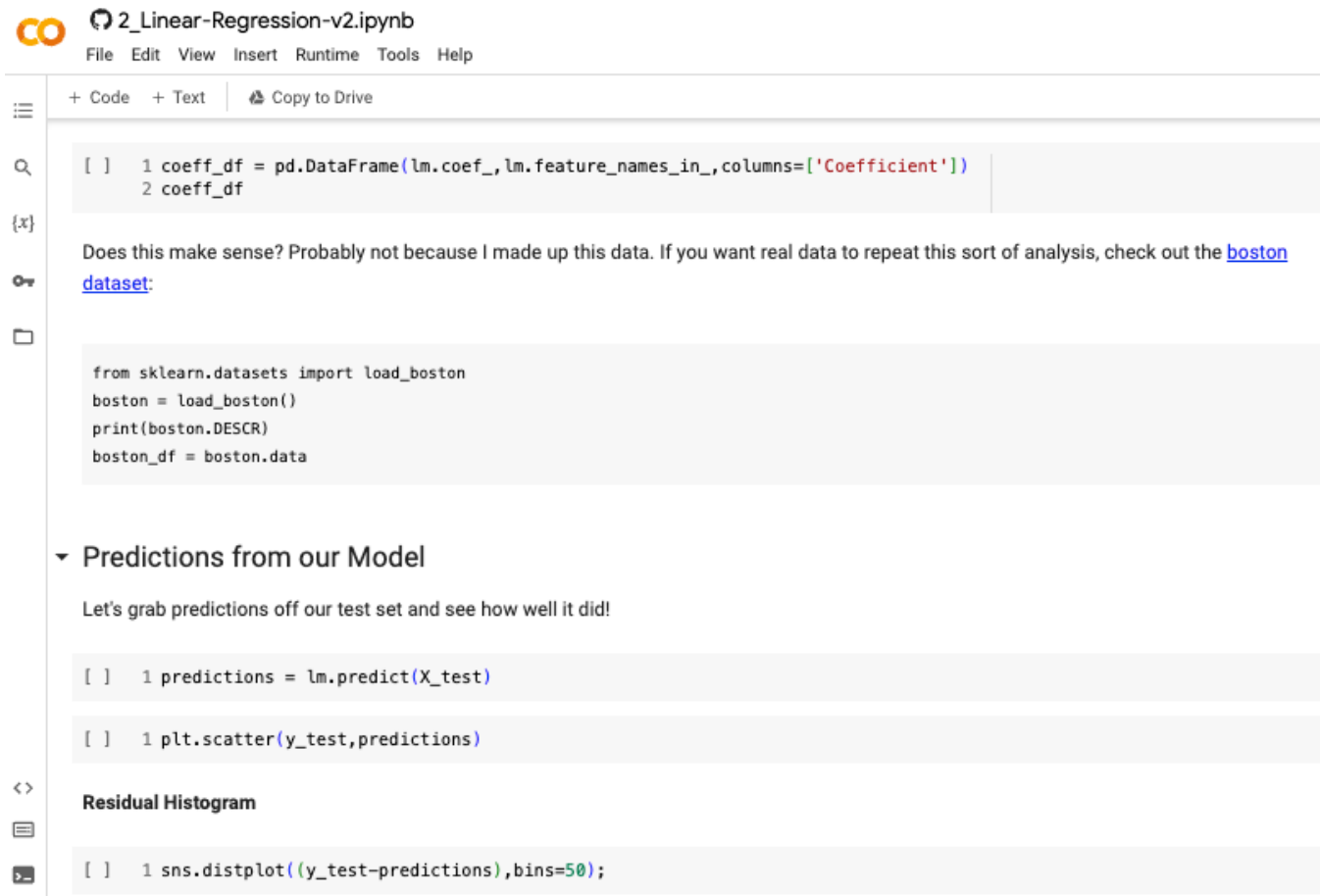
scgupta.me 
twitter.com/scgupta 
linkedin.com/in/scgupta 



Dev Module



Do you think that user can use this code to get the prediction result? **NO!!!**



co 2_Linear-Regression-v2.ipynb
File Edit View Insert Runtime Tools Help

+ Code + Text Copy to Drive

```
[ ] 1 coeff_df = pd.DataFrame(lm.coef_, lm.feature_names_in_, columns=['Coefficient'])  
    2 coeff_df
```

Does this make sense? Probably not because I made up this data. If you want real data to repeat this sort of analysis, check out the [boston dataset](#):

```
from sklearn.datasets import load_boston  
boston = load_boston()  
print(boston.DESCR)  
boston_df = boston.data
```

▼ Predictions from our Model

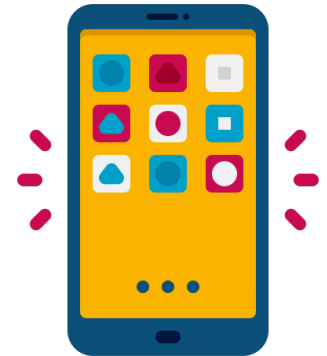
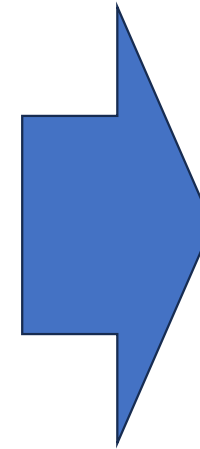
Let's grab predictions off our test set and see how well it did!

```
[ ] 1 predictions = lm.predict(X_test)
```

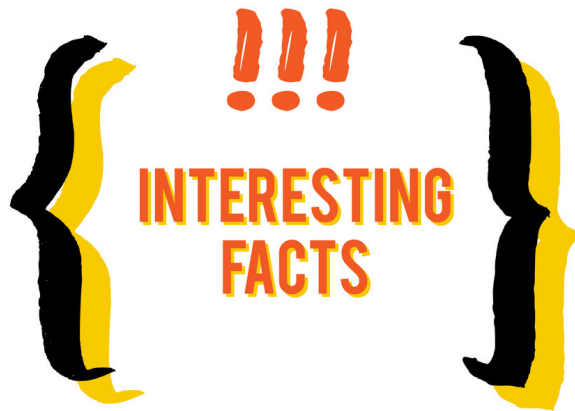
```
[ ] 1 plt.scatter(y_test, predictions)
```

Residual Histogram

```
[ ] 1 sns.distplot((y_test-predictions), bins=50);
```



Interesting facts



- More than 50% of AI projects were **FAILED** since they didn't plan about the deployment.

For the AI project, finish building the model is only 50% of the work.

The remaining work is about the deployment as a touchpoint to the target user.

Guideline for the dev module



- More than 50% of AI projects were **FAILED** since they didn't plan about the deployment.
- 1) Understand who is the target user first? E.g., customer vs staff
- 2) Action: Knowledge without action is meaningless. (API & Integration)
- 3) User journey: simple & easy-to-use
- 4) Model is already used via the touchpoint. Then, what next?
- 5) Always take administrative works into account

1) Understand who is **the target user** first?
E.g. customer vs staff

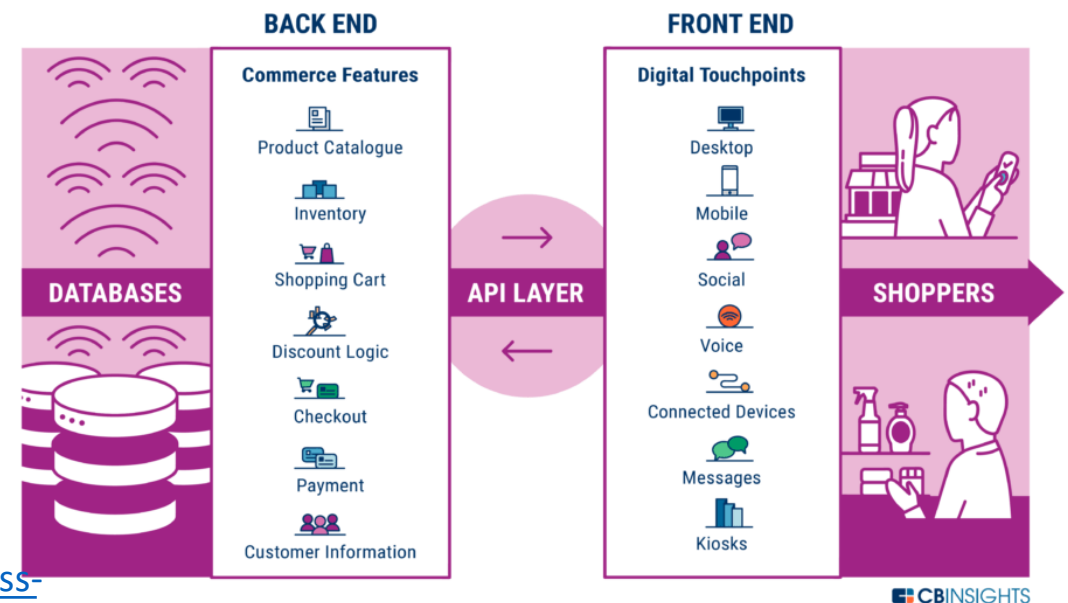


2) Action: Knowledge without action is meaningless. (API & Integration)

- API's for omnichannel
- Campaign automation
- Loyalty management

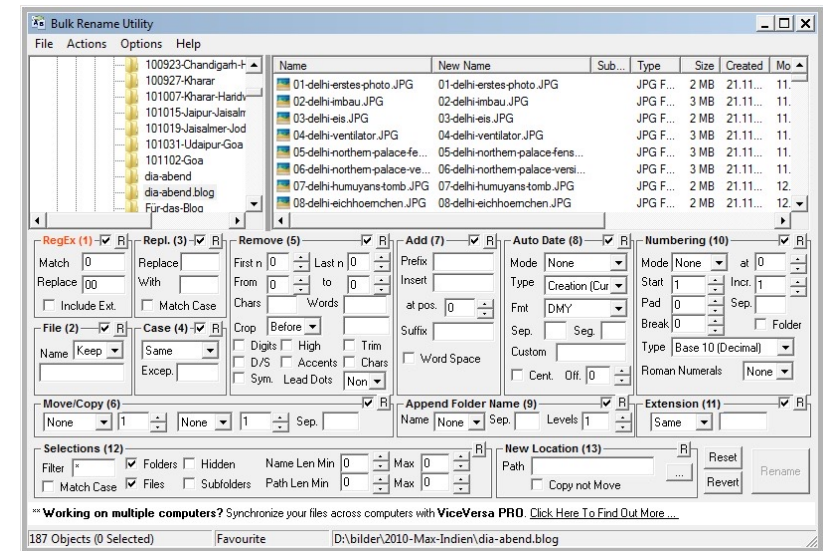
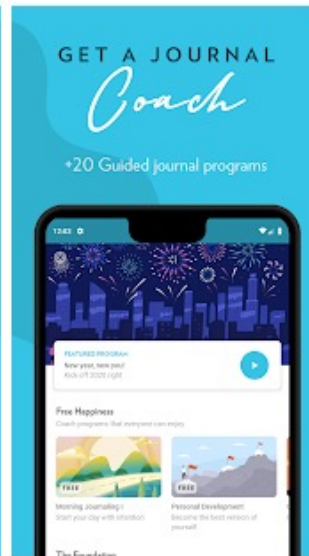
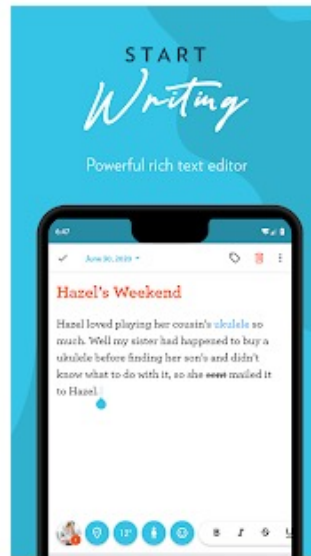
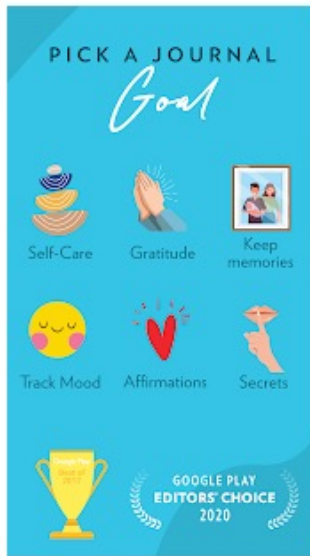
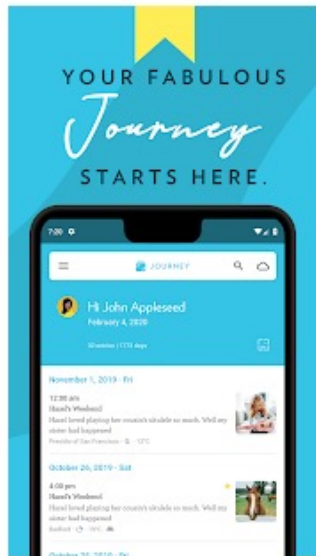


Components of a headless commerce architecture



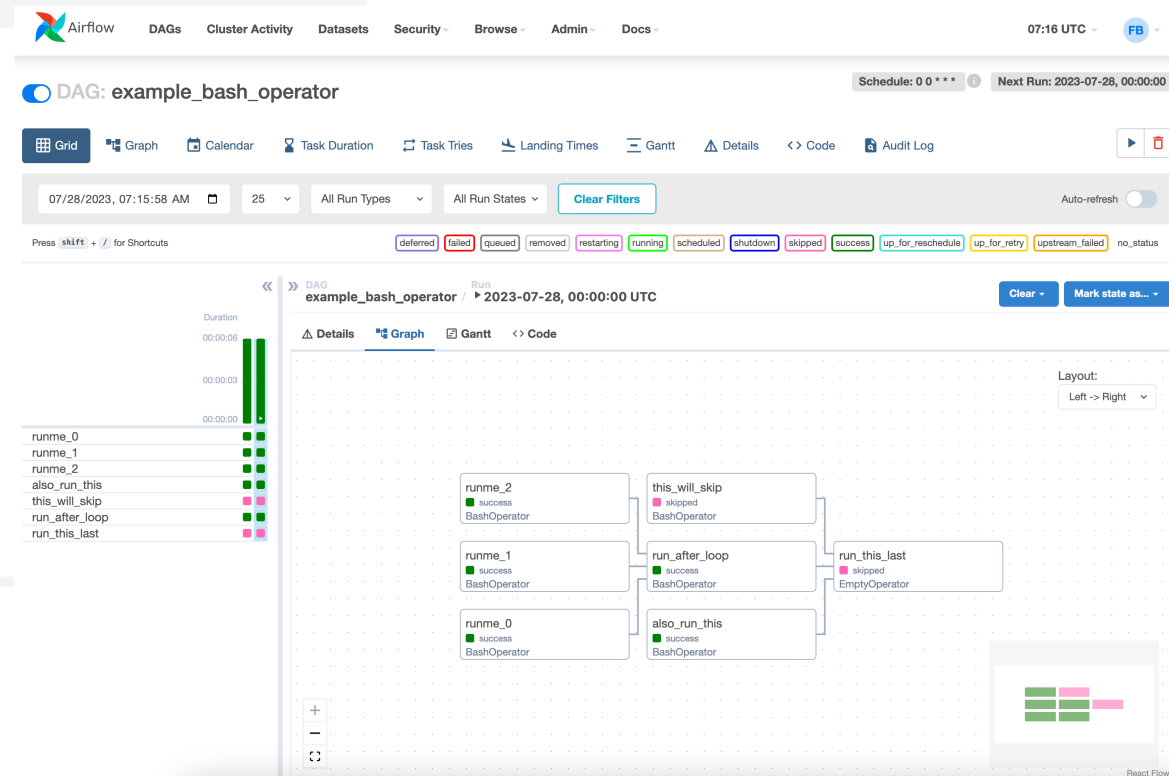
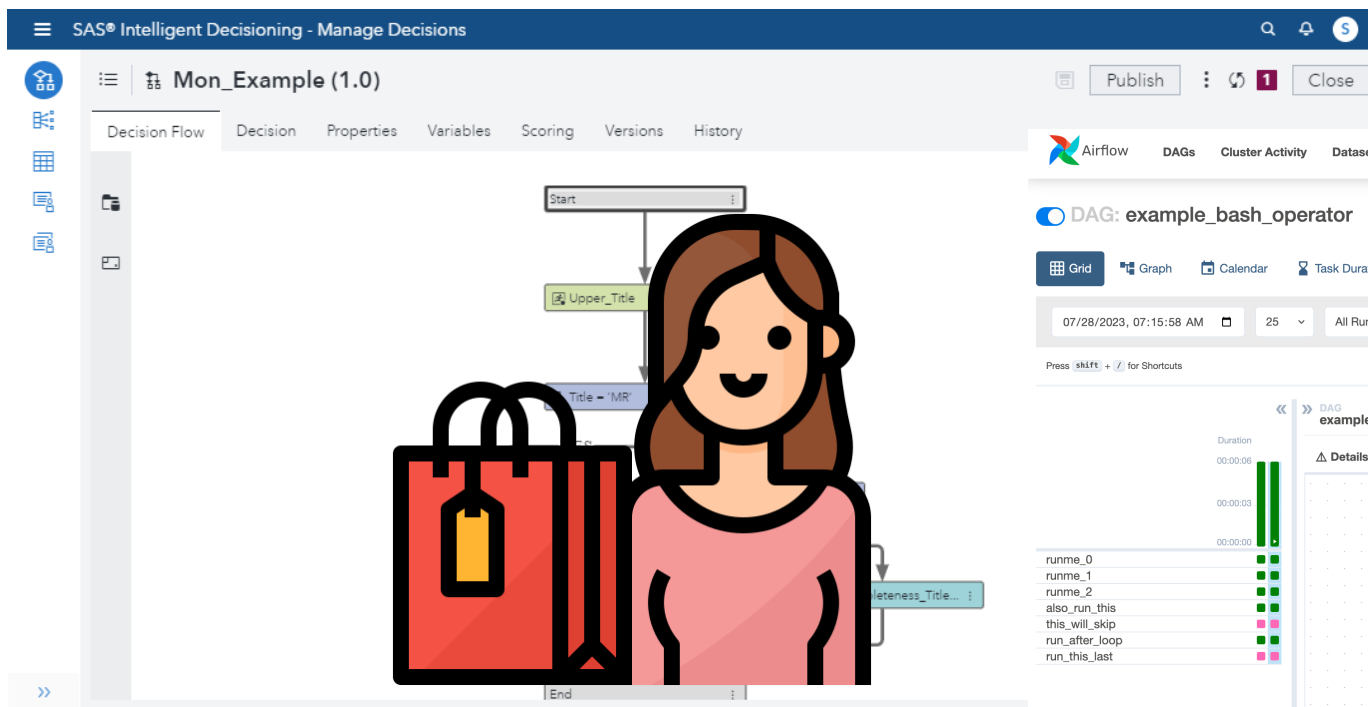
<https://www.cbinsights.com/research/report/what-is-headless-commerce/>

3) User journey: simple & easy-to-use



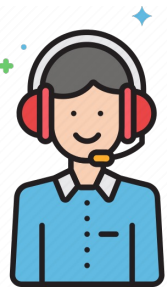
4) Workflow management

Model is already used via the touchpoint. Then, what next?



5) Don't forget about the staffs

Always take administrative works into account



Select a Campaign Starter Kit

Filter By Campaign Type: All

- All
- Build your own
- Renewal
- WinBack
- Event
- Welcome
- Fundraising

Blank Campaign
The blank campaign presents an empty canvas ready for your marketing automation artistry.

Basic Renewal Campaign
Standard renewal beginning at 90 days prior to expiration.

Win Back Campaign - 3-6 Months Lapsed
The goal of the Win Back campaign is to reach out to prior members to generate interest and encourage them to rejoin your organization.

Advanced Renewal Campaign
Advanced renewal beginning at 120 days prior to expiration, targeting inactive and active members.

Small/Mid Size Event Promotion Campaign
Promote webinars, local chapter events, and professional development courses. 3 messages over time.

Large Event Promotion: Open Registration/Last Chance
7 Messages to promote and drive event registrations.

xtreme

Steave Jobs. steave@gmail.com

Create New

Dashboard

Profile

Tables

Icon

Blank

404

Upgrade to Pro

Dashboard

Home Library

Sales Summary
Overview of Latest Month

20K

10K

0K

1 2 3 4 5 6 7 8

Feeds

- You have 4 pending tasks. Just Now
- Server #1 overloaded. 2 Hours ago
- New order received. 33 May
- New user registered. 30 May

Top Selling Products
Overview of Top Selling Items.

Monthly

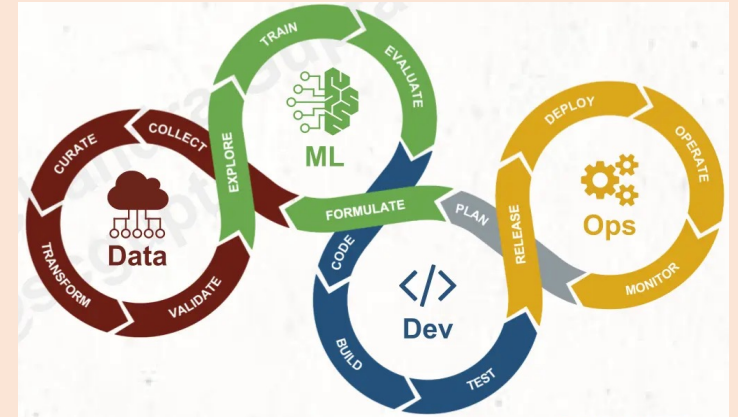
Products	License	Support Agent	Technology	Tickets	Sales	Earnings
EA Elite Admin	Single Use	John Doe	Angular	46	356	\$2850.06
MA Monster Admin	Single Use	Veronica Fern	Vue.js	46	356	\$2850.06
MP Material Pro Admin	Single Use	John Doe	Bootstrap	46	356	\$2850.06
AA Ample Admin	Single Use	John Doe	React	46	356	\$2850.06

Guideline for the dev module (recap)



- More than 50% of AI projects were **FAILED** since they didn't plan about the deployment.
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Ops Module



Guideline for the ops module



- MLOps (Ops = Operations)
- Release (versioning), configure, monitor
- 1) Monitoring: operation and performance
- 2) Model is not stationary and always needed to be retrained.
- 3) There are many MLOps tools nowadays.
- 4) BI Dashboard can help to summarize the collected data.

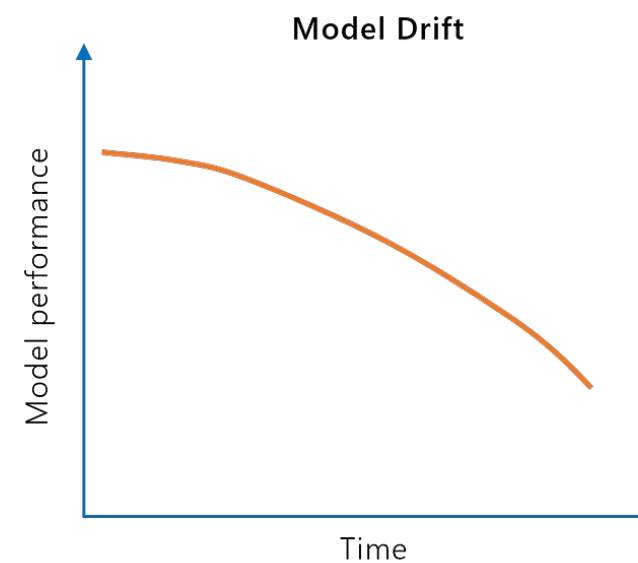
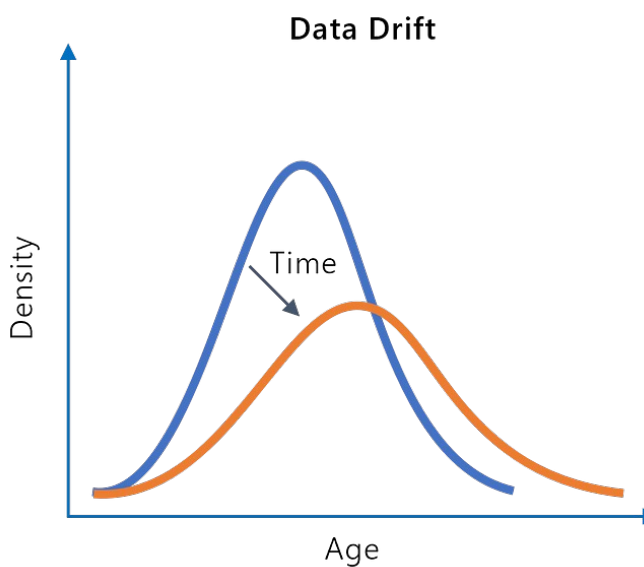


1) Monitoring

- Operation monitoring



- Performance monitoring



2) Model is not stationary and always needed to be retrained.



3) Many MLOps tools nowadays

Release (versioning), configure, monitor



mlflow

Tracking

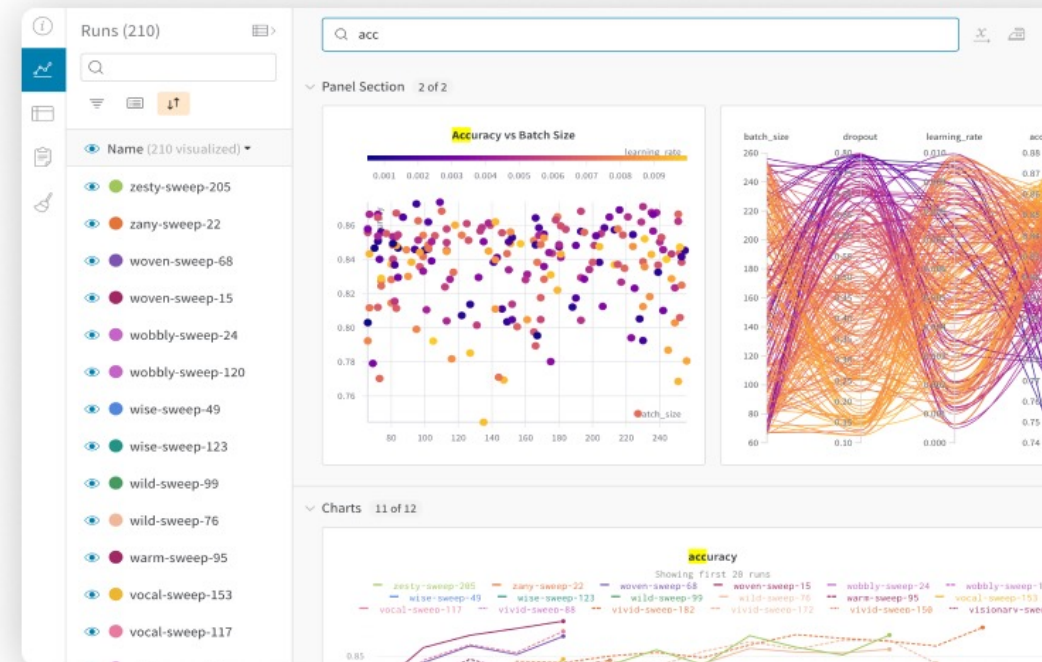
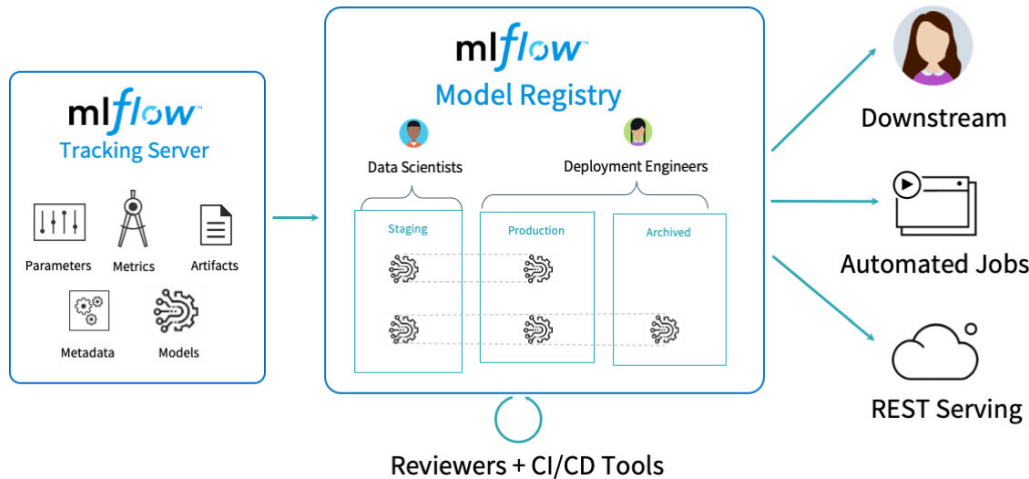
Record and query experiments: code, data, config, results

Projects

Packaging format for reproducible runs on any platform

Models

General format for sending models to diverse deploy tools



4) BI Dashboard can help to summarize the collected data.



Guideline for the ops module (recap)

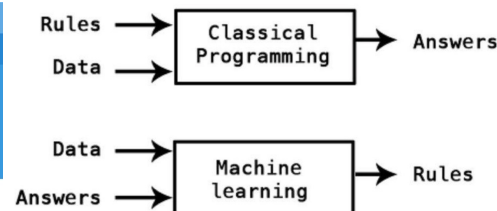
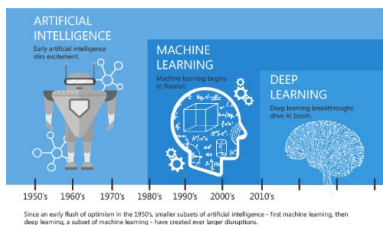
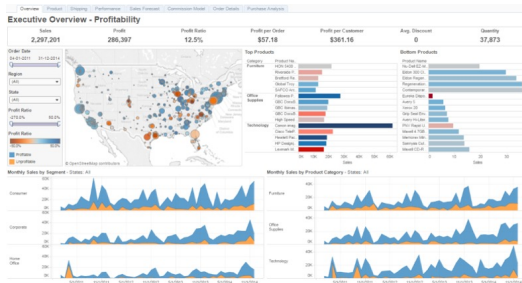


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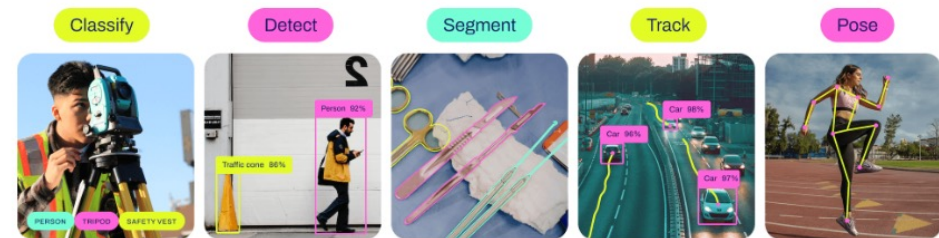
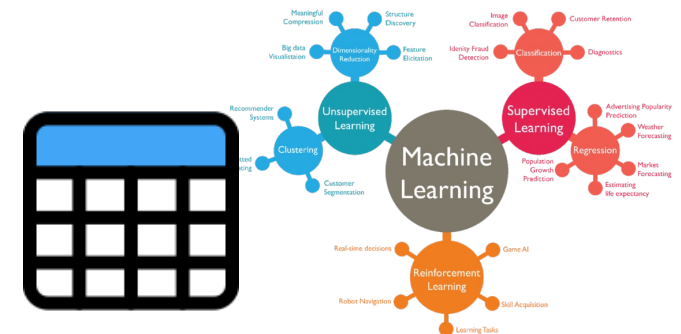
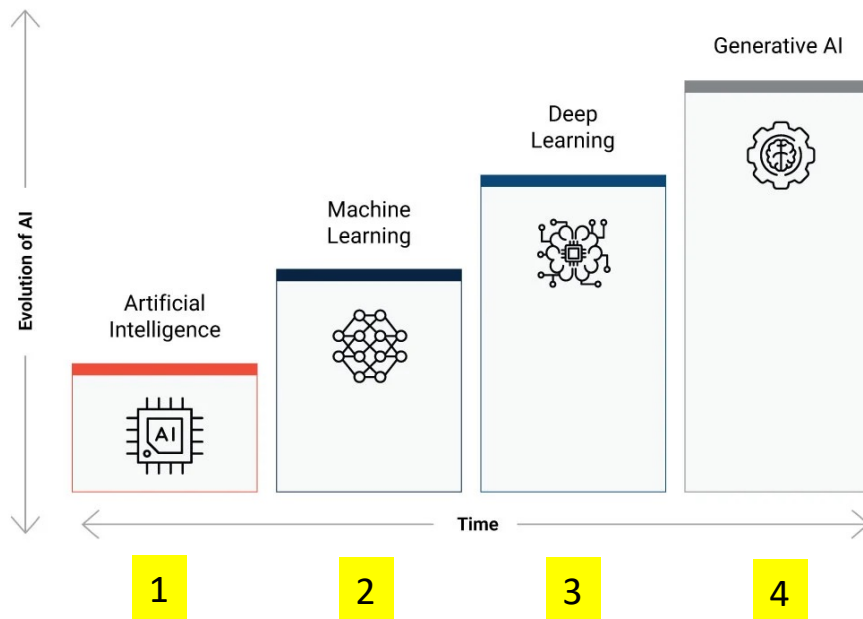
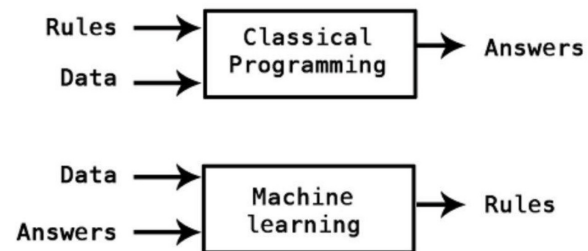
Conclusion

Remark: Choose the right tools 😊

NOT all problems **can or should** be solved by analytics!



Be able to understand about recent digital technologies (AI)



All [Models](#) download automatically from the latest Ultralytics [release](#) on first use.

OpenThaiGPT 1.0.0-beta

OpenThaiGPT Version 1.0.0-beta is a Thai language 7B-parameter LLaMA v2 Chat model finetuned to follow Thai translated instructions and extend 24,554 Thai words vocabularies for turbo speed. For more information, please visit [the project's website](#) | [Github](#).

Examples

ลดความซับซ้อนของคำอย่างไร

วางแผนเที่ยวในภูเก็ตแบบบริษัททัวร์

เขียนบทความ

เขียนโค้ด

คำนวณคณิตศาสตร์

แปลภาษา

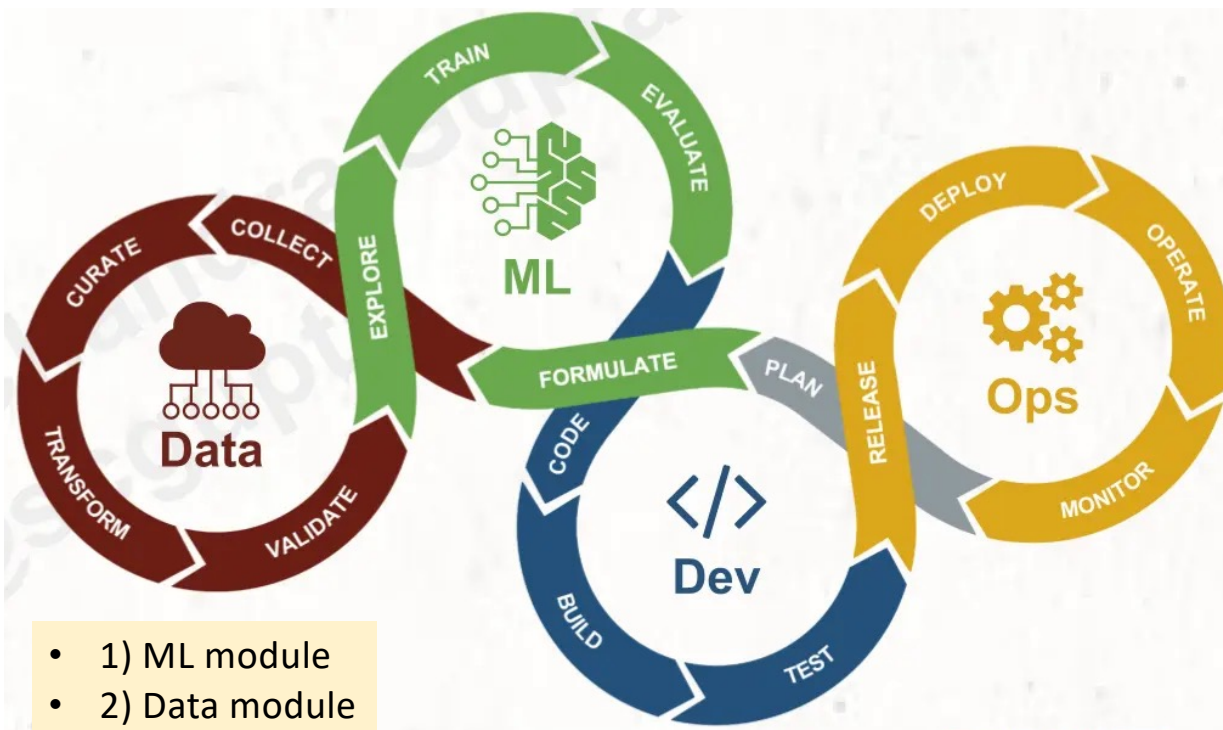
Instruction

แนะนำวิธีการหาเงินออนไลน์

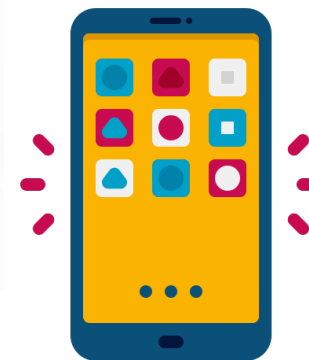
Input

คำถาม (ไม่จำเป็น)

Be able to understand and use the process of applying these technologies



- 1) ML module
- 2) Data module
- 3) Dev module
- 4) Ops module





THANK YOU
FOR YOUR
ATTENTION

Peerapon.v@chula.ac.th